Hybrid Forecast of Phytoplankton in the California Current

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Abstract: Most of the oxygen breathed by humanity is produced by phytoplankton. Like many things on Earth, these organisms are severely affected by climate change. We apply Machine Learning (ML) to the long-term data set provided by CalCOFI to analyze the influence of different environmental parameters on phytoplankton. For forecasting of phytoplankton - up to the year 2100 for various climate change scenarios - we propose a hybrid approach that combines ML models and physical climate models (CMIP6 models). In our work, we show that the best ML model is able to estimate the phytoplankton population in the California Current with high accuracy. This model is then used in conjunction with the CMIP6 climate models, to predict the development of the organisms for the far future and different climate scenarios. Our hybrid approach is capable of integrating the results of complex climate models into a forecast without increasing the computational effort. It offers the possibility of predicting phytoplankton based on different scenarios without running simulations.

Keywords: Forecasting, Phytoplankton, Climate Change

1 Introduction

This research examines the influence of climate change on phytoplankton in the California Current. We analyze which environmental parameters have a large influence on the phytoplankton population. Afterwards, these parameters are used to train a regression model that estimates the phytoplankton population with high accuracy. Finally, this ML model is combined with physical climate models to predict phytoplankton up to 2100.

Industry, agriculture, transportation, living standards, land use, all these are anthropogenic activities that have a significant influence on climate change [De23]. It is essential to understand the consequences of climate change and therefore to be able to take appropriate measures to mitigate its effects [Sá21]. Scientists agree that the key to this is to research the oceans [Sá21; Wo21]. These large and complex ecosystems cover 71 percent of the Earth's surface and are home to the majority of all living organisms on the planet [Um09].

The regulation of climate, temperature, carbon dioxide and oxygen levels, as well as its use as a food source, are only a fraction of the functions of this ecosystem. However, the oceans are increasingly suffering from intensive use by humans and from the effects of climate change. In addition to global warming and rising sea levels, the oceans are also acidifying. This has serious consequences for humans as well as for all other species. [Um09]

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Micro- and macroorganisms play a fundamental role in the oceans [Sá20]. This includes, for example, diatoms and flagellates, which are divided into zooplankton and phytoplankton [Um09]. Zooplankton refers to animal organisms that spend their entire life as plankton or are born only in the larval stage. Phytoplankton, on the other hand, refers to unicellular plant organisms. Through photosynthesis, these organisms produce most of the oxygen breathed by humanity [Na23]. For this reason, these species form the basis for a healthy and diverse life on Earth.

One of the four most important eastern boundary upwelling systems (EBUS) in the world is the California Current. EBUS are characterized by equatorward currents transporting cool, fresh and oxygen- and nutrient-rich water along the west coast of a continent. These wind-driven upwelling systems ensure that nutrients reach the surface, making these regions highly productive and very important for fisheries. In summary, the California Current constitutes a highly relevant ecosystem for the planet. Combined with the abundance of freely available data, research in this region contributes significantly to advancing our understanding of complex marine ecosystems and their responses to climate change. [Th22]

2 Background

Related Work Studying the oceanic systems is an essential part of climate research. Observing the marine ecosystem as well as its micro- and macroorganisms plays an important role in analyzing climate change [Wo21]. According to Sánchez-Pi et al. [Sá20], these organisms are key to understanding the complex functions of ecosystems and the risks posed by climate change. Meanwhile, scientists are using ML methods to explore the oceans and predict and mitigate the consequences of climate and biodiversity changes [Sá20].

Righetti et al. [Ri19] use an ML approach to make predictions about biographic patterns of phytoplankton species. The data used are based on 180 000 locations distributed globally. Additionally, different phytoplankton species are distinguished to investigate diversity. They tested three ML methods to train a classification model, which informs about the presence or absence of a species. Due to its high degree of interpretability and flexibility, they selected the generalized additive model (GAM). [Ri19]

In addition to machine learning methods, which primarily learn relationships within the training data to predict a target variable, environmental parameters can also be forecasted using physical climate models. These models simulate physical and chemical processes, as well as the dynamic interactions between different atmospheric and oceanic components, in order to produce projections for the 21st century and beyond or to reconstruct past states [KB24]. However, such simulations are highly computationally intensive [No18]. For instance, the computations for the CMIP6 models are largely carried out on high-performance computing systems, such as the Mistral supercomputer operated by the German Climate Computing Center [Gm]. Moreover, several years are typically required for model development and the completion of simulations [Ey16]. Therefore, it can be assumed that individual climate model simulations may take several weeks, months, or even years to complete.

Particularly for research questions at the local scale or for target variables that are not explicitly contained by climate models, a hybrid modeling approach can be advantageous. This concept is implemented, for example, in Nowack et al. [No18], where existing outputs from a climate model (e.g., temperature) are used as input features in a regression framework to estimate another parameter (e.g., ozone concentration).

A comparable approach is presented by Rivera et al. [Ri23], who not only investigate how plankton abundance can be modeled using machine learning methods, but also assess how this abundance may change under future climate conditions. First, a classification model is trained to determine the geographic distribution of zooplankton in the Humboldt current system by calculating the presence or absence of phytoplankton per pixel. Then, two Representative Concentration Pathways (RCP) scenarios are used to determine the values of the oceanographic variables for the periods 2040-2050 and 2090-2100. Lastly, the trained model can be projected onto the future with the help of these values. In contrast to [Ri23], our approach learns a regression model that predicts the exact chlorophyll-A concentration for the CalCOFI measuring stations.

In summary, previous work has mainly focused on classification approaches, which indicate only the presence or absence of phytoplankton, without providing information on its actual abundance. Many machine learning models simply extend past trends, often failing to capture unexpected shifts in climate dynamics, such as those driven by political decisions. Scenario-based analyses offer the advantage of examining whether phytoplankton is influenced by climate change or develops independently of it. While physical climate models simulate various environmental parameters and their interactions, they require significant computational resources. Therefore, there is a need for more efficient approaches that utilize existing data without relying on costly simulations. Future research should address the following questions: Which parameters influence phytoplankton abundance? How is it linked to climate change? And what forecasts can be made under different SSP scenarios up to 2100?

Data The data for this research is provided by CalCOFI[2], this marine ecosystem research program is one of the longest-running sampling programs in the world. In general, it studies the physics, biochemistry and biology of the marine environment. Data collection started in the late 1940s and was initiated to investigate relationships between oceanographic conditions and sardine abundance. These goals and the volume of data associated with them quickly expanded. The data set contains 113 defined stations distributed along the California coast from the northern part of San Francisco Bay to San Diego on a uniform grid, with stations placed near the coast to approximately 670 km offshore (Fig. 1). The research ships collect samples from each station for 2 to 4 weeks every quarter of the year

[2] California Cooperative Oceanic Fisheries Investigation: https://calcofi.org/

using Conductivity-Temperature-Depth (CTD) rosettes at different depths. Thus, profiles are created from the sea surface to the bottom or to a maximum depth of 515 m. [Sa21]

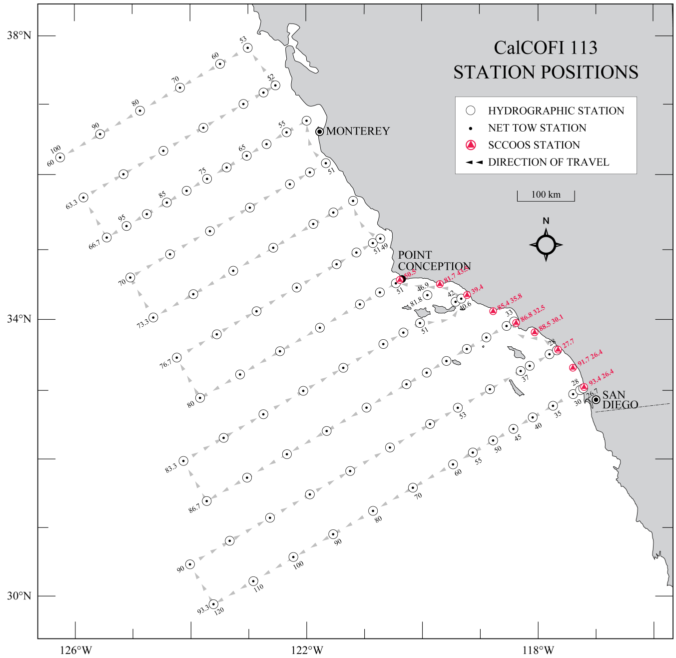

Fig. 1: Map of the CalCOFI stations.[3] [Ca23]

In addition to the oceanographic data used to train the ML model, we used estimations from climate models, including various Shared Socioeconomic Pathways (SSP) scenarios to be able to predict into the future. For this task, we chose a high-quality data set provided by Bio-ORACLE v3.0[4]. It contains comprehensive raster data sets on the sea surface and benthos [Bi24]. These data include geophysical, biotic and ecological information [Bi24] for the present as well as for the future, based on CMIP6 models [As24]. The data products of the sixth phase of CMIP are considered an important source of robust and reliable climate information [Gm].

The Bio-ORACLE layers are global data sets that can be used for both large- and small-scale research due to its high spatial resolution. The spatial resolution is displayed in 5 arcmin, which corresponds to approximately 9.2 km at the equator [As18]. The calculation of the Bio-ORACLE layer is based on averaging the results of the different CMIP6 models [As18]. All the data used in our research therefore use the same combination of climate models.

[3] SCCOOS = Southern California Coastal Ocean Observation System

[4] Bio-Oracle: https://www.bio-oracle.org/index.php

3 Methodology

Our hybrid approach consists of two main parts (Fig. 2): First, an explainable regression model is trained using the CalCOFI data, then in the second part, this model is used to create a forecast for phytoplankton occurrence for the years 2030 to 2100.

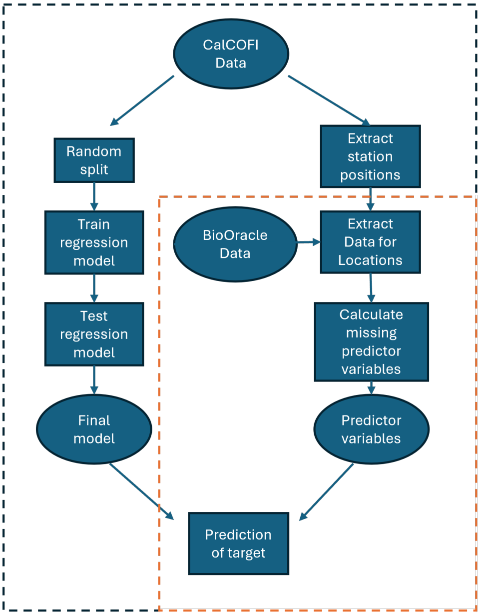

Fig. 2: Flowchart of the Workflow.

Chlorophyll-A was selected as the target variable for the regression model. Chlorophyll-A is a pigment found in all phytoplankton species [Da18] and is widely used as a standard proxy to estimate phytoplankton biomass [Je80]. The unit of chlorophyll-A is µg/l. An initial examination of the dataset revealed that chlorophyll-A concentrations exhibit a highly skewed distribution, which complicates both the interpretation of temporal trends and the analysis of relationships between chlorophyll-A and environmental variables. In cases where the data approximate a log-normal distribution, as observed here, a commonly applied approach is to perform a logarithmic transformation [Fe14; Ze21]. This transformation adjusts the relative differences between values, thus modifying the distributional characteristics of the variable [Ze21]. The transformation is applied using the formula: $x' = \log(x + 0.001)$

To ensure a good interpretation, only algorithms that guarantee a high level of interpretability were tested. This means that the flexibility of these algorithms is limited and, consequently, no complex models are generated [Ja13]. For this reason, linear regression (LR), generalized additive models (GAM) and random forest (RF) are tested. LR is considered the simplest and fastest approach, giving the user a good overview of the parameters that have the strongest influence on the target variable [Ja13]. Furthermore, the LR can be used as a baseline. GAM allows non-linear relationships to be represented, which often increases the accuracy of the predictions and the results are still easy to interpret [Ja13]. Furthermore, this method is used in some comparable works, such as that of Rivera et al. [Ri23]. RF combines a high degree of flexibility and still provides some interpretability due to feature importance (FI) scores [Ja13]. All future values of environmental parameters fall within the range of the training data, validating the applicability of the RF model for projection.

The models were trained and evaluated using an 80/20 train-test split. To compare the performance of the different algorithms, evaluation metrics such as R^2, root mean squared error (RMSE) and mean absolute error (MAE) were calculated (table 1). Among the tested models, RF achieved the best performance across all evaluation metrics. Consequently, the RF model was selected for the prediction of the occurrence of phytoplankton.

Tab. 1: Comparison R^2, RMSE and MAE of the LR-, GAM- and RF-Model based on the same training data.

Error Metrics	LR	GAM	RF
R^2	0.7696	0.8390	**0.9242**
RMSE	0.9037	0.7651	**0.5151**
MAE	0.6766	0.5684	**0.3264**

A correlation analysis was carried out before selecting the input parameters. This showed that there are few parameters that exhibit collinearity. Examples of these are the parameters density, water temperature and sampling depth. In general, RF is robust against correlations between features, and forecasts are not affected by this. For this reason, it was experimentally tested which input characteristics lead to the lowest error rates (the same evaluation metrics as in table 1). The input parameters selected are listed in Table 2.

The aim of the second part of our approach is to predict phytoplankton for different climate change scenarios as well as for different time periods in the future (2030-2100). The workflow for this part is visualized in the orange section in fig. 2. The basis for this are the Bio-Oracle data, which contain global raster data sets for various environmental parameters. Based on the geographical position of the measuring stations, the corresponding values for different future points in time can be extracted from the raster data sets and converted into tabular format. The format of this must correspond to that of the regression model's training data. Unfortunately, not all environmental parameters used in the regression model are covered by Bio-Oracle, requiring the missing parameters to be estimated for future points in time. Depending on the parameter, the mean or median (from the CalCOFI data) is used. As only a small subset of parameters was affected (nitrite, density & wind conditions), they were not excluded from the analysis to assess their relevance for chlorophyll-A and thus improve the interpretability of the prediction quality. Once the data set is generated, the final regression model can be applied to create the long-term forecast, i.e., predict the chlorophyll-A value until 2100. In order to interpret the model predictions, they must first be transformed back, as they are still subject to the logarithmic transformation.

4 Results and Discussion

The use of the explainable ML method RF enables the interpretation of the influence of individual environmental parameters on the target variable by calculating the feature importance (FI) scores (Tab. 2). These indicate that sampling depth, water density and water temperature have a significant influence on the abundance of phytoplankton (FI 29%-13%). Nutrients such as nitrate, silicate and phosphate as well as parameters related to position (distance to the coast and bottom depth) have a medium influence on phytoplankton with a FI <10%. Atmospheric parameters such as air temperature, air pressure and wind conditions have a FI of <2% and therefore do not have a major influence.

Tab. 2: Feature Importance score for all environmental parameters of the final regression model.

Feature Importance of environmental parameters			
Parameter	FI (%)	Parameter	FI (%)
Sampling depth	29.36	Phosphate	3.48
Water temperature	18.13	Salinity	2.14
Density	13.96	Air temperature	1.32
Nitrate	7.49	Quarter	1.11
Silicate	6.49	Pressure	0.99
Nitrite	5.69	Wind speed	0.78
Distance	4.66	Wind direction	0.70
Bottom depth	3.70		

The focus of this research was to compare different climate change scenarios to analyze whether the development of phytoplankton is affected by climate change. When inspecting the chlorophyll-A predictions over time and stations (Fig. 3) , we notice that each scenario leads to a reduction in phytoplankton occurrence . This trend is already evident in past data measured by CalCOFI. However, the extent of the loss varies greatly depending on the scenario considered. The sustainable SSP-scenarios SSP1-1.9 and SSP1-2.6 show a loss up to 2040 and change only slightly thereafter. Predictions of Chlorophyll-A concentrations decrease more steeply for scenarios that expect a high increase in CO_2 emissions until 2100. It can be seen that the higher the expected CO_2 increase, the lower the predicted chlorophyll-A concentration in 2100.

The main reason for the lower concentration of chlorophyll-A is the increase in water temperature. As already mentioned, the water temperature correlates significantly with the chlorophyll-A concentration. Bio-Oracle data show that the increase in CO_2 will also have a significant impact on the water temperature. Fig. 4 shows the predicted water temperature for different SSP scenarios as modeled by Bio-Oracle, which also shows an upward trend for nonsustainable scenarios. An increase in the water temperature leads to a higher grazing rate of zooplankton, exceeding the production of phytoplankton [Ol13]. Furthermore, warming leads to a reduction in the supply of nutrients. There is no exchange of water layers, which means that nutrients from the depths no longer reach the surface [Ho11].

These developments have a significant impact on this ecosystem and humanity. A reduced phytoplankton population leads to a loss of ecological resilience [He21], as well as to a negative development in the productivity of the marine ecosystem [He21; Ho11; Ol13]. This reduction in productivity affects fisheries, meaning that an important food source disappears [Fr21]. Phytoplankton also plays a crucial role in the global carbon cycle. If phytoplankton declines, the ocean loses its ability to absorb CO_2, which in turn further exacerbates climate change [Ho11].

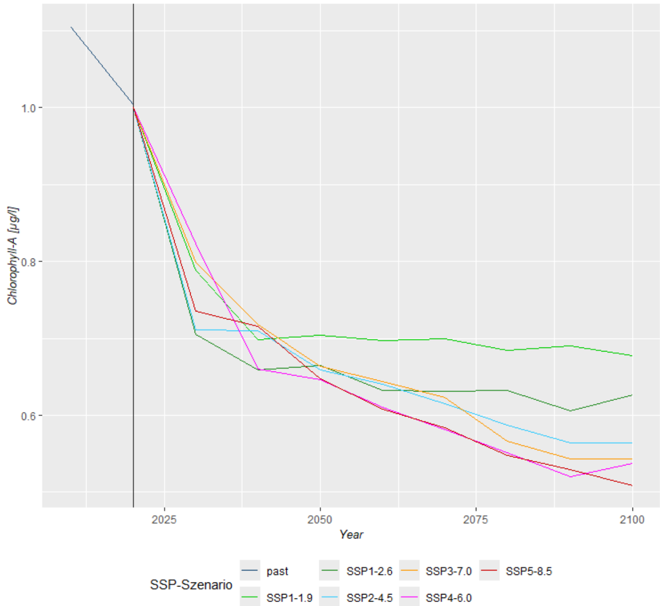

Fig. 3: Phytoplankton (chlorophyl-A) forecast

When reviewing the quality and plausibility of these results, two aspects must be considered. First, the quality of the regression model's ability to depict the relationship between various environmental parameters and the target variable is crucial. In our work, the RF model achieved an R^2 of 0.92, indicating good quality. Secondly, the Bio-Oracle data only contain predicted values, meaning an unknown level of uncertainty is included in the calculation. However, as Bio-Oracle combines several complex climate models from CMIP6 [As18], it can be assumed to be of suitable quality for further processing.

This hybrid forecasting approach offers several advantages over using either ML or physical climate models alone. Using only ML could lead to very inaccurate long-term predictions, as these would only be based on the distant past. However, the climate is constantly changing, which can lead to changes in the trend or unexpected fluctuations. Furthermore, various political decisions, as well as other events, can significantly influence climate development. Therefore, it makes sense to integrate different scenarios, like the SSP scenarios, into the analysis when creating such forecasts.

The simulation of such scenarios is usually carried out with atmosphere-ocean general circulation models (AOGCM). Atmospheric and oceanic models are combined to simulate the dynamics and interactions between the systems [KB24]. However, the calculations of these models are extremely complex and require a great amount of computing power and

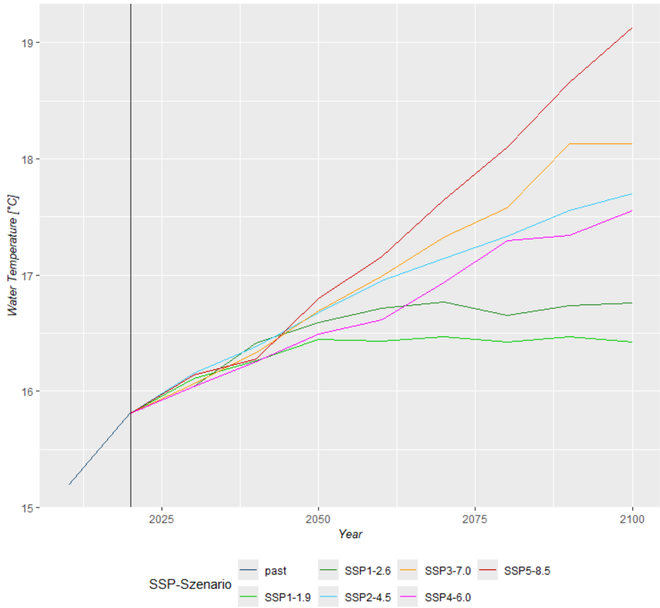

Fig. 4: Water temperature forecast

time [KB24]. In contrast, our approach uses fast ML methods to analyze relationships between environmental parameters and phytoplankton. The values already calculated by various AOGCMs can then be used as input for the forecasts. This allows the heterogeneous correlations of the simulations to be integrated into the prediction without having to execute complex calculations. This approach is therefore ideal to create what-if scenarios.

5 Conclusion

This research shows that explainable ML algorithms perform well in representing the correlation between different environmental parameters and phytoplankton. In particular, the RF approach achieves more accurate prediction results than GAM, which is typically used in this area.

To summarize the results, all SSP-scenarios lead to a reduction in phytoplankton occurrence. However the extent of the loss varies greatly depending on the scenario. It can be noted, that the higher the expected CO2 increase, the lower the predicted chlorophyll-A concentration in 2100. This variation in predictions depending on the scenarios supports the hypothesis that phytoplankton could be affected by climate change. According to the feature importance of the regression model, the main factors responsible for phytoplankton development are water temperature, sampling depth, and density. Furthermore, the influence of nutrients and geographical conditions can also be observed.

Overall, our hybrid approach is able to integrate results of physical climate models into a forecast without increasing the computational effort or calculation time. Furthermore, this approach offers the possibility to calculate the development of phytoplankton depending on different scenarios without running simulations. In this paper, the analysis is focused on the average development over time in the different scenarios, future research could explore spatial differences in phytoplankton occurrence and development.

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